



AROUND THE WORLD

- ▶ Ashok Jhunjhunwala's corDECT WiLL solution has already been deployed in several countries: Madagascar, Brazil, Fiji, Nigeria, Iran, Egypt, Argentina, Kenya, Yemen, Tunisia, Angola and Nepal.
- ▶ In Brazil, the "Computador Popular," including the monitor, has an estimated price tag of \$250 (about Rs 11,200). It was built by academics at the behest of the government, which is anxious to help low-income Brazilians span the Digital Divide. Sergio Vale Aguiar Campos, a professor at south-western Brazil's Federal University of Minas Gerais, said that the Brazilian government had basically asked, "How low can you go?"
- ▶ In China, basic desktop computers powered by V-Dragon CPUs and sufficient to browse the Internet would cost as little as \$200. Culturecom Holdings Ltd says its V-Dragon CPU retails for only \$15-\$30. Co-developed by IBM and based on the Midori Linux OS, the V-Dragon architecture is aimed at the Greater China market. Culturecom is hoping that its CPUs will help to bridge the Digital Divide in China.

we seem to be getting a growing number of computers, and using them highly inefficiently. One example is the Cyberage scheme in Goa, where more than 17,000 students got computers—which are used largely highly inefficiently."

If Dr Ashok Jhunjhunwala of IIT-Madras had his way, we would see computing based on the STD booth model. Jhunjhunwala's goal is to providing Internet connectivity and telephony on a commercial basis to some 600,000 villages across India. Jhunjhunwala has long been working with his TeNeT group of IIT-Madras in Chennai to work out the technologies and economic models that make this feasible and sustainable.

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Dr Ashok Jhunjhunwala,
Professor, Dept of
Electrical Engineering,
IIT-Madras



Dr Ashok Jhunjhunwala of IIT-Madras has a vision for computing in India based on the STD-booth model

mobile and two million Internet connections. The Internet is power, and the country needs 200 million telephone and Internet connections in its cities, small towns and rural areas," argues Jhunjhunwala. And, he says, subsidies are not going to achieve this.

"At the prevalent rates, telecom is not affordable to more than two or three per cent of households in India." Jhunjhunwala's solution is to reduce costs. This, he believes, would take phones and the Internet to India's approximately 650,000 villages and 2,000 very small towns.

State-run BSNL already has 25,000 rural exchanges, of which 70 per cent are connected on fibre. If this is added to a wireless system with a 10-km range, it would cover over 80 per cent of villages in India, he argues. Given India's population density, this could offer one Net connec-

tion for every 500 people, and one to two connections per village.

If optimally utilised, the cost of Jhunjhunwala's corDECT wireless in local loop (corDECT WiLL) solution makes costs plunge to Rs 10,000. This brings in one 35/70 Kbps Internet connection plus a simultaneous telephone link. If the area is sparsely populated, the cost goes up to Rs 14,000.

"Today, more than 25 per cent of India's total telephone revenues comes from STD booths, and almost 30 per cent of the population has started to use telephones," says Jhunjhunwala. He says that the Internet "enables people. It is changing the way we live... those without the Internet will have a tremendous disadvantage as we go on. People with confidence and enabled people can make all the difference. We would like to see that all villages get reasonable speed Internet connection at the earliest."

The Price Conundrum

Coming back to the question of whether a \$70 computer is possible: (also see box "Ravi Pradhan: The Mouthpiece of the Masses?," page 55) Sanjeev Keskara of AMD India is of the opinion that computers have already almost reached their lowest possible price points, and that a further reduction in costs is not probable or feasible. He says, "The CPU contributes 10 to 15 per cent of the overall desktop. There are other key components such as the motherboard, hard disk, monitor, RAM, CD-ROM... which are the major building blocks for desktops. We have seen for many years that technology is changing very fast in all areas of the desktop building blocks, and that the customer is getting superior technology products; but it is hardly seen that we start getting old technology at lower prices.

"For example, if you used to get a 20 GB hard disk at price x two years ago, last year you got a 40 GB at a similar price point, and tomorrow you will get 80 GB at the price point of today's 40 GB. But you will not get a 20 GB at a fourth of the price of



A BASIC PC: HOW MUCH?

Since we have been talking about price, let's put things into perspective. Forget about gamers and graphic designers for a moment. Also forget about flaunt value. How much does the hardware in India cost now, for a computer that most people, including office and home users, need? Here's a reckoner.

Processor: AMD Sempron 2400+, Rs 3,250
Motherboard with onboard LAN card: Asus KM400, Rs 3,250
RAM: 128 MB DDR-333, Rs 1,100
Hard disk: Seagate / Samsung 80 GB, Rs 3,000
Mouse: Logitech first mouse, Rs 300
Keyboard: Logitech New Touch, Rs 275
Cabinet: HiS, Rs 1,000
CD-ROM: Sony, Rs 750
Floppy drive: Sony, Rs 350
Speakers: Creative SBS15, Rs 450
Monitor: Samsung 15-inch, Rs 4,700
Internal Modem: Rs 450
Total: Rs 18,875